USN | | | | | | | | | | [22CSBS208]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

CONVERSATIONAL SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Tell the Importance of NLU with suitable example.	L1	CO2	**02**
	b.	List out the common applications of Natural Language Processing.	L1	CO2	**02**
	c.	Explain Rule Based Chatbot and mention its disadvantages.	L2	CO2	**02**
	d.	Define "modes of engagement" in the context of human interaction.	L1	CO1	**02**
	e.	Define dialogue Management.	L1	CO3	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain different messaging platforms with suitable examples.	L2	CO1	**05**
		OR			
	b.	Discuss the security issues in Conversation System.	L2	CO1	**05**
3.	**a.**	Explain the Architecture of Chatbot.	L2	CO2	**05**
		OR			
	b.	Explain the positive and negative impact of Chatbot on Business.	L2	CO2	**05**
4.	**a.**	Discuss the importance of POS tagging? explain with suitable examples	L2	CO2	**05**
		OR			
	b.	Explain the popular data bases in Conversation system.	L2	CO2	**05**
5.	**a.**	Describe the essential framework to construct chatbot.	L2	CO3	**05**
		OR			
	b.	Explain Intent Classification Techniques	L2	CO3	**05**

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6.	**a.**	Explain the role of Natural Language Processing (NLP) in conversational systems?	L2	CO4	**05**
		OR			
	b.	Describe the importance of Information Extraction in conversational systems with suitable example.	L2	CO4	**05**
7.	**a.**	Define two key conversational metrics used in contact centers. Explain their importance in evaluating the quality and effectiveness of customer interactions.	L2	CO5	**05**
		OR			
	b.	Explain how to generate text from the NLG database with programming Example.	L2	CO5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Explain the Phases of NLP with suitable examples.	L2	CO1	**10**
		OR			
	b.	Discuss the history and impact of AI in detail.	L2	CO1	**10**
9.	**a.**	Discuss all possible underlying Technologies used in Chatbot.	L2	CO2	**10**
		OR			
	b.	Design a python program for sentiment analysis.	L6	CO2	**10**
10.	**a.**	Articulate how NLP resolves the word sense Disambiguates (WSD). Write a python program to demonstrate the same.	L6	CO2	**10**
		OR			
	b.	Compare NLP, NLG and NLU? Explain by taking suitable example.	L4	CO2	**10**
11.	**a.**	Discuss different Dialogue management in detail with example of each.	L2	CO3	**10**
		OR			
	b.	Design a Rule based Chatbot using python.	L6	CO3	**10**

12.	**a.**	Explain the role of Automatic Speech Recognition (ASR) in conversational systems. How do machine learning models process speech to generate accurate text transcriptions, and what challenges do they face (e.g., accents, background noise)?	L2	CO4	**10**
		OR			
	b.	Discuss the role of information extraction in conversational systems. How does machine learning enable the extraction of structured data from unstructured user inputs.	L2	CO4	**10**
13.	**a.**	Describe the concept of advanced dialog management in conversational systems. How do machine learning-based dialog policies differ from rule-based approaches?	L2	CO5	**10**
		OR			
	b.	Define A/B testing. Discuss different types of A/B Testing.	L2	CO5	**10**